

Project Proposal & Technical Report

Trusted Government Knowledge, Rules & Document Assistant

A Data-Driven Generative AI & Document Scrutiny Platform for the Government of Assam

Problem Statement No.	PS No. 46 · NESFIC-D-36
Submitted to	Administrative Reforms, Assam Administrative Staff College (AASC), Department of Science & Technology, Pension & Public Grievances Department, and Assam State Space Application Centre (ASSAC), Government of Assam
Submission type	Functional Deep-Tech MVP Prototype · Demonstration-Ready
Applicant	Globizhub India Private Limited
Date	October 2026

Notice & Disclaimer: This document represents an independent technical proposal and functional prototype submitted in response to challenge problem statement PS No. 46 / NESFIC-D-36 issued under the North East Seva First Innovation Challenge 2026 (NESFIC 2026) under the Seva Sankalp Abhiyan. It is designed to demonstrate technical feasibility, deterministic zero-hallucination statutory verification, and decision-support workflows. It is not an officially commissioned, endorsed, or operational system of the Government of Assam or any department thereof.

1. Company Overview & Corporate Identity

Company Name	Globizhub India Private Limited
Industry Sector	Information Technology (IT) Services & Software Solutions
Website	https://globizhub.com/

2. Registration & Accreditation Details

Corporate Identification Number (CIN)	U74999KA2019PTC120377
DPIIT Recognition Number	DIPP250200
MASI Registration Number	MASI2025/1029

3. Office Locations & Addresses

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4. Leadership & Key Contacts

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5. Problem Statement

Administrative governance in Assam is operationalised through a large portfolio of gazettes, notifications, service rules, executive guidelines, and office memoranda across multiple nodal line departments — Administrative Reforms & Training, Assam Administrative Staff College (AASC), Department of Science & Technology (DST), Pension & Public Grievances Department, and Assam State Space Application Centre (ASSAC). Each department manages critical statutory mandates, qualifying service determinations, and citizen-facing services. Yet an operational gap exists between legislative intent and administrative disposal:

1. For Department Desk Officers & Secretariat Branches: Official rules, amendments, and executive circulars reside scattered across scanned historical PDF archives, legacy paper records, and disparate departmental desks. Desk officers spend hours manually searching for prevailing clauses and verifying whether an earlier rule has been superseded by a subsequent circular, leading to file pendency and procedural delays.

2. For State Administrators & Civil Service Training (AASC): When inducting and training Assam Civil Services (ACS) and departmental personnel, curricula rely on static digests that quickly fall out of sync with real-time statutory amendments. Supervisory authorities lack a centralized, searchable intelligence platform to monitor knowledge utilization and ensure uniformity across departments.

3. For Technical & Spatial Governance (DST & ASSAC): Complex technical guidelines for remote sensing, geospatial land demarcation, and digital infrastructure require strict adherence to statutory specifications. Non-technical officers frequently struggle to locate and interpret specialized norms without inter-departmental referrals.

4. For Retiring Employees & Public Pensioners: Retiring government servants and citizens face significant bureaucratic friction understanding pension eligibility, qualifying service calculations, and requisite forms (e.g., Form 7, No Demand Certificates) under the Assam Services (Pension) Rules 1969, often resulting in avoidable grievances.

6. Solution Overview

We present Trusted Government Knowledge, Rules & Document Assistant — a full-stack governance and legal intelligence platform that transitions administrative knowledge management from periodic manual search to continuous operational intelligence. The platform comprises five interconnected operational layers:

1. Dual-Layer 300 DPI OCR & Ingestion Pipeline: High-throughput document processor featuring adaptive contrast enhancement, skew correction, and dual-layer layout analysis. Extracts text layers from both clean digital gazettes and degraded historical scanned circulars, with dedicated support for English and Assamese statutory fonts.

2. Structural Chunking & Statutory Rule Parsing Layer: Custom algorithmic chunker that detects statutory section boundaries (e.g., Rule 41(2), Section 9(1)) and consolidates structured lists, preventing broken clauses and preserving legal context.

3. Deterministic Retrieval & Substantive Quote Extractor: Hybrid search engine pairing BM25 keyword matching with dense semantic embeddings. A substantive sentence extractor isolates operative legal provisions to eliminate decorative boilerplate.

4. Zero-Hallucination Strict Verifier Guardrail: Mathematical character-level NFKC string grounding validator. Every statement generated by the assistant is cross-referenced against the verbatim text layer of the cited gazette. If a claim lacks exact substring grounding, the response is rejected with a safe refusal.

5. Secretariat Green-Sheet Noting & Cryptographic Audit Layer: Formats verified statutory answers directly into standard Assam Secretariat Manual notings. Every ingested document chunk and generated noting is timestamped and cryptographically hashed (SHA-256), establishing a continuous audit trail.

7. Core Architectural Capabilities

The platform introduces the following architectural innovations tailored to Assam's administrative framework:

1. Character-Level NFKC String Grounding (Zero Hallucination Guarantee): The verifier normalizes unicode characters and verifies that every statutory citation is an exact, byte-level substring of the official gazette. Extrapolations are strictly prohibited.

2. Role-Based Access Control with Server-Side Enforcement: Dedicated workspaces for STATE_ADMIN (full statewide visibility), DEPT_OFFICER (department-scoped with instant noting generation), and CITIZEN (plain-language eligibility summaries with transparent document page viewers).
3. Safe Refusal & Audit Escalation Protocol: When queried on matters outside the active statutory index, the platform refuses to synthesize ungrounded speculation, instead logging the query for departmental nodal officer review.
4. Bilingual Script Ingestion Readiness: Foundational tokenization pipeline architected for seamless processing of Assamese script () alongside English administrative gazettes.
5. Sovereign On-Premise Deployability: Designed for deployment within the Assam State Data Centre (SDC) or MeitY-empanelled sovereign cloud infrastructure, ensuring confidential government files remain within state-controlled perimeters.

8. Technology Stack & Implementation Status

Implemented Prototype Features

Frontend Framework	Next.js 16 (App Router, Turbopack), React 19, TypeScript 5, Tailwind CSS 4, shadcn/ui, Lucide React icons, Framer Motion
Accessibility & UI	GIGW 3.0 compliant, Screen Reader ARIA landmarks, A-/A+ font resizer, SpeechSynthesis audio narrator (Hear), Bilingual selector (EN/AS)
Backend & Database	FastAPI / Python 3.9+ asynchronous REST API, SQLite3 relational metadata index, PyMuPDF (fitz) dual-layer OCR extraction engine
Grounding & RAG Engine	Hybrid BM25 + dense vector indexing, deterministic substantive quote extractor, character-level NFKC verifier engine
Access Control	Role-scoped session handling (Super Admin, Desk Officer, Citizen), scoped document filtering via dept_scope
Audit Ledger	SHA-256 cryptographic document chunk hashing, immutable noting timestamps, CSV telemetry export

Proposed Production Architecture

Direct Gazette Ingestion	Automated sync connector with the Assam Government e-Gazette repository and departmental MIS portals
Enterprise IAM	Integration with official State SSO (Jan Parichay / e-Pramaan) with Multi-Factor Authentication (MFA)
Sovereign Deep Tech SLM	Fine-tuned 7B/14B parameter Indian Legal Small Language Model deployed on dedicated SDC GPU nodes (vLLM / TensorRT-LLM)
Bilingual Assamese OCR	Custom vision-language OCR pipeline fine-tuned on historical Assam Government font ligatures and stamp marks
Security & Compliance	Full CERT-In empanelled VAPT auditing, STQC GIGW 3.0 compliance certification, and DPDP Act 2023 adherence
Workflow Automation	Integration with e-Office (NIC) for direct insertion of AI-generated verified green-sheet notings into official files

9. Current Stage & Demonstration

Functional MVP Prototype · Demonstration-Ready. A working full-stack prototype is deployed using official Assam gazettes across Pension Rules 1969, ARTPS Act 2012, Mission Basundhara guidelines, and DST circulars. The MVP includes multi-route navigation, role-based desk isolation, dual-layer OCR extraction, character-level verification, automated green-sheet noting generation, and a continuous audit trail.

Demonstration URL: <https://nesfic-d-36.vercel.app> (Demonstration environment)

Source Code Repository: <https://github.com/anupam-codespace/NESFIC-D-36.git>

Feature	Status	Production Scope
Multi-Route Navigation	Implemented	Integration with State portal master navigation
Accessibility Suite (TTS & Font Resizer)	Implemented	STQC certified WCAG 2.1 AAA compliance
Multi-Department Gazette Ingestion	Implemented	Automated e-Gazette webhook listener
Dual-Layer 300 DPI OCR Extraction	Implemented	Distributed GPU-accelerated OCR workers
Character-Level Verifier Guardrail	Implemented	Formally verified mathematical proof engine
Secretariat Green-Sheet Noting Generator	Implemented	Direct integration with NIC e-Office file dispatch
Role-Based Desk Isolation (Admin/Officer/Citizen)	Implemented	Jan Parichay enterprise SSO integration
Live Telemetry & Performance Benchmarks	Implemented	Real-time Prometheus/Grafana monitoring
SHA-256 Cryptographic Audit Ledger	Implemented	Hyperledger / State Blockchain notarization
Public Transparency & Citizen Inquiry	Implemented	Sewa Setu / RTPS citizen service integration

10. Stakeholder & Context Analysis

Target Stakeholders

1. Primary Regulatory & Administrative Bodies (B2G): Administrative Reforms Department (statutory compliance), Assam Administrative Staff College (ACS civil service training), Department of Science & Technology & ASSAC (geospatial/technical guidelines), and Pension & Public Grievances Department.
2. Internal Administrative Cadre: Joint Secretaries, Deputy Secretaries, Under Secretaries, Section Officers, and Senior Assistants across line departments responsible for file drafting and statutory approvals.
3. Public Beneficiaries: Retiring government employees, pensioners, and citizens who benefit from transparent entitlement interpretations and verified eligibility calculations.

Contextual Differentiation

Existing administrative workflows rely on manual circular physical files and static digests. Generic AI and ticketing platforms lack awareness of Assam's statutory workflows, secretariat green-sheet noting hierarchies, and departmental jurisdictions. VidhiAI is purpose-built for the Assam context — it models the exact statutory relationships between principal acts and amending notifications, ensures 100% evidence-grounded answers, and formats results directly into the green-sheet notings officers already know.

11. Intellectual Property & Licensing

- **IP Status:** Proprietary software architecture developed for the challenge; the source code for the demonstration prototype is provided for evaluation purposes.
- **Patent Strategy:** Provisional domestic patent planned covering the method and architecture for real-time, zero-hallucination statutory verification, automated green-sheet noting generation, and cryptographic provenance tracking in administrative governance.
- **Deployment Model:** Proposed as an indigenous SaaS / GovTech model deployable on the State Data Centre (SDC) or MeitY-empanelled cloud infrastructure, with the option of an on-premise deployment for air-gapped secretariats.

12. Funding & Staged Milestones

Requested Grant: ₹40,00,000 (Rupees Forty Lakhs) — Deep Tech Grant / Assam Startup Scheme.

#	Milestone	Budget	Share	Prerequisites	Deliverables
1	Multi-Department Data & Gazette Corpus Engineering	₹6,40,000	16%	Gazette archives from AASC, Pension, ARTPS, DST, ASSAC	Comprehensive Assam statutory ontology; structured JSON schema; multimodal ingestion pipeline
2	Bilingual Assamese Vision-Language OCR & Ingestion Engine	₹10,40,000	26%	Historical gazettes corpuses; State Data Centre sandbox	Custom OCR fine-tuned on Assamese ligatures (98%+ benchmark); 300 DPI dual-layer connector with retry logic
3	Sovereign Legal SLM & Zero-Hallucination Grounding Engine	₹10,80,000	27%	Secretariat Manual precedents; annotated green-sheet notings	Domain-adapted 7B/14B parameter Legal SLM; character-level NFKC verifier engine; automated statutory conflict graph
4	Multi-Department Secretariat Pilot & Security Hardening	₹7,60,000	19%	Pilot clearance (Administrative Reforms, AASC, Pension); MeitY staging cloud	60-day live pilot across 3 nodal departments (100+ officers); CERT-In VAPT audit report; GIGW 3.0 accessibility verification
5	Legal & Scale Readiness, Provenance Ledger & SDC Deployment	₹4,80,000	12%	Pilot telemetry data; SDC production container clearance	Evaluation report; statewide rollout blueprint; SHA-256 cryptographic provenance ledger; SDC bare-metal deployment
Total		₹40,00,000	100%		

13. Projected Impact & Evaluation

Targeted estimates based on pilot benchmarks.

Administrative Impacts

1. **Accelerated Evidence Retrieval:** Reduces evidence-assembly and rule cross-referencing time for file reviews from hours to minutes, dramatically accelerating Secretariat decision-making.
2. **Proactive Operational Triage:** Surfaces operative clauses, qualifying service rules, and entitlement ceilings within seconds of query submission — eliminating the lag between file intake and supervisory disposal.

3. Objective Merit Recognition: Continuous per-scheme progress and audit-logged noting records establish an objective basis for recognizing high-performing officers and departments.
4. Citizen-Centric Transparency: Plain-language entitlement summaries and transparent source document inspections restore citizen trust in government pension and public service delivery.

Employment & Capacity Targets

- Direct Technical Roles: 6–10 engineering, AI/ML research, legal ontology, and GIS data roles based in Assam (Guwahati branch office).
- Ecosystem Support Roles: 25–40 indirect technical and training support positions across districts (nodal officers, digitization operators, field trainers).

14. Alignment with Assam Startup Policy 2025–2030

Primary Category: Category B — IT, ITeS & Artificial Intelligence (Deep Tech Specialization)

The proposal aligns with the "Innovate Assam 2030" key strategic focus areas:

1. Domestic GovTech Innovation: Indigenous technology built for Government of Assam needs, addressing the specific statutory workflows of Administrative Reforms, AASC, DST, Pension & PG, and ASSAC.
2. Data Governance: High-performance web tools, zero-hallucination verification algorithms, RBAC, and audit trails engineered by regional talent based at the Guwahati branch office.
3. Citizen-Centric Governance: Supports the statutory objectives of transparent pension processing, entitlement verification, and grievance redressal under the Assam Right to Public Services Act (ARTPS).
4. Employment Generation: Direct and indirect technical roles based in Assam, with specialized AI training programs for secretariat staff and departmental nodal officers.

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